accumulate wealth that is 12.5 (= 50/4) times her final salary. As indicated before, I treat this as an example of traditional retirement planning advice.

Exhibit 5.9

Minimum Necessary Savings Rate to Accumulate 12.5× Final Salary

Maximum Sustainable Withdrawal Rates

For 50/50 Asset Allocation, 30-Year Work Period, 30-Year Retirement Period

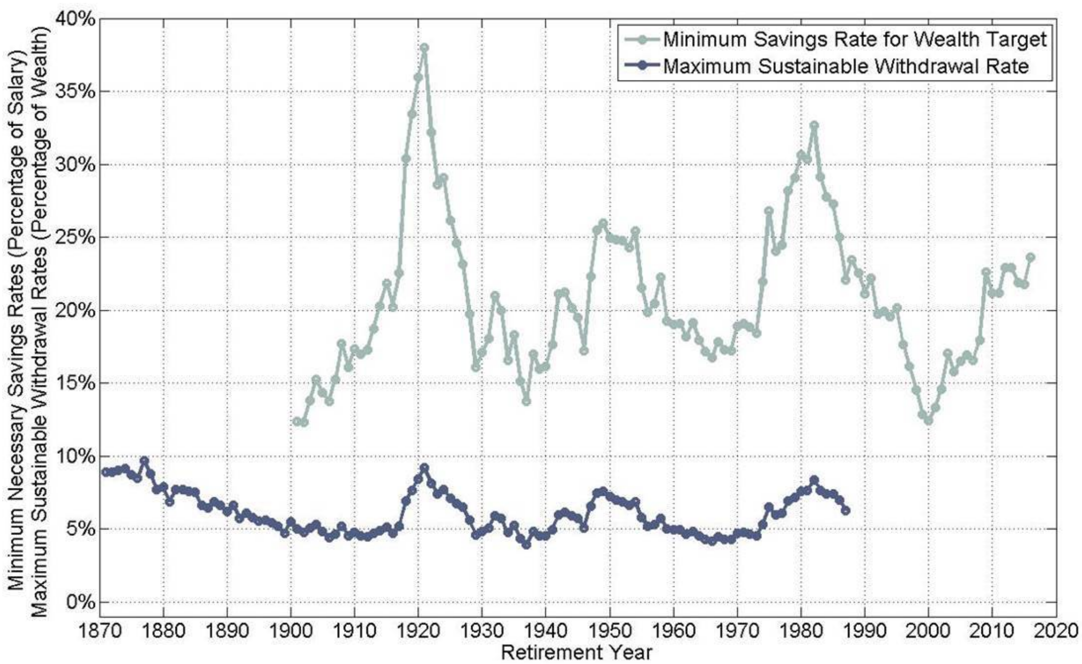

Exhibit 5.9 shows the path of minimum necessary savings rates (MSRs) for thirty-year careers to achieve the fixed wealth accumulation goal. These MSRs are volatile and can be quite high. They were over 30 percent for new retirees between 1918 and 1922 and in 1980–1982. Interestingly, the pattern of these MSRs closely follows that of the corresponding maximum sustainable withdrawal rates (MWRs) for each retirement year. For instance, the largest MSR of 38 percent occurs for someone retiring in 1921, the same year we see the largest MWR of 9.2 percent. In addition, retirement years that experienced relative lows for MWRs also experienced relative lows for MSRs. The MSR for 1929 was 16.1 percent, 13.8 percent for 1937, and 16.7 percent for 1966. These retirement years experienced MWRs of 4.6 percent,